Subhanjal Pant

+977 984-8813316 | 78bme043.subhanjal@pcampus.edu.np | Github Profile Link

Education

Bachelor in Mechanical Engineering May 2021 – May 2026
Pulchowk Campus, Institute of Engineering (IOE), Tribhuvan University *Kathmandu, Nepal*

- **Key Coursework:** Engineering Drawing, Thermodynamics, C programming, Basic electronics, Basic Electrical Engineering, AutoCAD, Engineering Mechanics, Metrology, Strength of Materials, Control System, Finite Element Method, MATLAB, Maple

TECHNICAL SKILLS

- **Engineering Design:** SolidWorks (CAD), ANSYS (Static Structural)
- **Fabrication:** Machining, Welding, Cutting, Grinding, 3d Printing
- **Electronics:** Arduino, ESP32, Sensor calibration, Point-to-point Wiring, Mechatronics integration, Soldering
- **Robotics & State Estimation:** ROS2 Humble and Gazebo Classic, Nav2 Stack, MoveIt, Kalman Filters (KF, EKF, UKF), Sensor Fusion, State Estimation
- **Programming:** C, C++ and Python. (Github Link)
- **Data Analysis and Machine Learning:** NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, Classical ML (Regression, Clustering, KNN). (Github Link)
- **Computer Vision** OpenCv
- **Tools:** Comfortable with Linux, Git/Github, Docker, LaTeX

Familarity

- **Pytorch:** Working knowledge of Neural Networks
- **KiCad:** Basic circuit design and work flow

PROJECTS

1. **Autonomous Navigation with Differential Drive Robot**
   - Simulated a differential drive robot in ROS2 Humble and Gazebo Classic, implementing real-world kinematics and sensor integration.
   - Deployed autonomous navigation and occupancy-grid mapping using LiDAR data and the Nav2 stack; configured SLAM Toolbox for real-time localization.
   - Engineered custom publisher/subscriber nodes and managed ROS2–Gazebo bridge for sensor data integration.

   Github Repository   Video Link

2. **Pneumatic Height Control System (Final Year Project)**
   - Designed and fabricated a pneumatic testbed from scratch using welding, lathe machining, and structural assembly to evaluate non-linear control strategies for height regulation.
   - Integrated ESP32, VL53L0X ToF sensors, pressure sensors, and IMU; implemented onboard Kalman filtering for robust real-time state estimation.
   - Conducted comparative experimental evaluation of Bang-Bang, PID, and Fuzzy Logic controllers under variable load and impact disturbances.
   - Comparing traditional PID against Bang-Bang and Fuzzy Logic to determine robustness in variable height and impact scenarios.

   Technical Summary Report Link   Full Project Report Link   Video Demonstration Link

3. **Circuit Craft**
   - Designed and developed a multi-modular educational platform for secondary students, funded by the **Ministry of Education, Science and Technology, Nepal**.
   - Engineered five experimental stations, including a logic gate array for binary addition/subtraction and a Tesla Coil for wireless energy transmission.

- Consolidated diverse physical phenomena into a single interactive interface to facilitate experiential learning and visualize invisible scientific principles.

Report Summary Link

4. **Tic-Tac-Toe with computer mode (with Heuristic AI) and multi-player mode in C programming language**
   - Developed a 700+ line interactive game in C featuring a custom rule-based algorithm for automated gameplay.
   - Implemented look-ahead logic to prioritize winning moves and block opponent victory paths, moving beyond simple stochastic (random) move generation.
   - Developed programming logic, debugging and managing large codebase for the first time

   Github Link Video Link

HONORS AND ACTIVITIES

- **National Rank 188 (Top 1.25%):** Got 188 rank nationally among 15,000 in IOE entrance examination securing full merit based scholarship in Mechanical Engineering.
- Got admission in **Pulchowk Engineering Campus** reputed as the **top engineering campus** of Nepal
- Recipient of **ANSWER Nepal scholarship**, awarded to high academic potential students with weak economic background since grade 9.
- Took part in the **STEAM (Science Technology Education Arts and Mathematics)** competition conducted and a project sponsored by the **Ministry of Education, Science and Technology, Nepal.**

PROFESSIONAL EXPERIENCE

- Did 2 months internship at NAST (Nepal Academy of Science and Technology) - worked on design, modelling, fabrication and electronics integration of Tomato Picking Machine.
- Did 1 month Trainee Internship distributed along 4 months at NIC (National Innovation Center) - Learnt welding, electronics like using arduino, bluetooth module HC05, motors
- Worked in Robotic Club, Pulchowk Campus for a month - Learnt using Lathe, Drilling Machine, Grinding, Cutting